Apurva Kokate

Artificial Intelligence PhD Candidate, Oregon State University

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Areas of Expertise

AI driven drug discovery - Graph Neural Networks - Micro Structure exploration - Human centric AI systems

Education

Doctor of Philosophy, Artificial Intelligence *2021-Present*
Oregon State University, Corvallis, OR, USA

Master of Science, Computer Science *2016-2018*
Iowa State University, Ames, IA, USA

Bachelor of Engineering, Computer Engineering *2014-2018*
Vidyalankar Institute of Technology, Mumbai, Maharashtra, India

Selected Research & Publications

MOSE-GNN: A Motif-Based Self-Explaining Graph Neural Network for Molecular Property Prediction **(LoG conference, Poster)**
Location: Oregon State University *2024*

- Designed *MOSE-GNN*, a novel **self-explaining GNN architecture** that guides model training towards **chemist intuitive rational for molecular design**. link
- Developed a molecular fragment-based interpretability framework producing *chemically meaningful* and *global* explanations.
- Demonstrated superior fidelity, consistency, and motif-correctness over post-hoc explainers.
- Enabled chemist-aligned model transparency for safety-critical molecular property prediction tasks.

Interpretable Deep Learning for Guided Microstructure–Property Exploration in Photovoltaics **(npj Computational Materials, Article)**
Location: Iowa State University *2019*

- Applied interpretable deep learning to identify **microstructural attributes** most predictive of photovoltaic efficiency.
- Built **explainable CNN pipelines** enabling scientists to trust, validate, and utilize ML-driven insights in materials discovery.
- Integrated attribution analysis with physics-based reasoning to accelerate photovoltaic materials exploration. link

A Study of Interpretability Mechanisms for Deep Convolutional Networks **(Iowa State University Repository, Thesis)**
Location: Iowa State University *2018*

- Conducted a systematic study of **vision-based interpretability** algorithms, creating a taxonomy for pixel-level attribution.
- Implemented and benchmarked saliency, perturbation, and decomposition techniques across deep CNN architectures.
- Revealed strengths, limitations, and failure modes of popular interpretability approaches used in both academia and industry.

A Forward–Backward Approach for Visualizing Information Flow in Deep Networks **(NeurIPS, Full-paper)**
Location: Iowa State University *2017*

- Co-developed a **layer-wise attribution** method using mathematical inversion of the forward pass to analyze information flow.
- Improved interpretability and debugging of CNNs, yielding a documented **15% improvement** in classification accuracy.
- Demonstrated how forward–backward signal decomposition enhances transparency and reliability in deep neural networks.

Position: Explainable AI Needs Domain Knowledge Consensus to Be Useful **(OpenReview, Full-paper)**
Location: OpenReview *2025*

- Authored a position paper arguing that *domain knowledge consensus* is essential for achieving actionable and trustworthy XAI.
- Analyzed limitations of current model-agnostic explainability methods and highlighted failure cases arising from domain-agnostic interpretation.
- Proposed a framework recommending *domain-aligned priors*, evaluation protocols, and ontology-driven validation for reliable XAI.
- Presented guidelines advocating collaboration between ML researchers and domain experts to establish shared interpretability standards.

Work Experience

Graduate Assistant **Machine Learning**
Location: Oregon State University *2021–2025*

- TA for **AI 534 (Machine Learning)**, **CS 434 (Machine Learning & Data Mining)**, and **CS 325 (Analysis of Algorithms)** .
- Mentored students in algorithm design, data structures, supervised learning, model evaluation, and debugging best practices.

Full Stack Software / ML Engineer **Text Analytics**
Location: Kingland, Iowa *July2018-May2021*

- Co-created Kingland's enterprise Text Analytics platform, integrating **Deep-learning (NER), contextual language modeling, and document-parsing pipelines**
- Engineered cloud-native ingestion and NLP services that processed 5M+ financial records and 1.1M+ daily updates, enabling large-scale extraction and normalization of entities from highly variable unstructured content.
- Achieved 99% entity-resolution accuracy for DTCC's mutual-fund data by designing context-aware extraction workflows addressing challenges such as alias resolution, domain-specific language, and bias mitigation referenced in "Managing Bias" and "Understanding Context" blog posts.
- Delivered measurable business value by automating manual document review, improving data quality, and enabling risk monitoring and compliance workflows described in Kingland's "Text Analytics Use Cases" series.
- Built end-to-end features across the full stack — microservices, APIs, data pipelines, and UI components — ensuring seamless integration of ML models into production systems serving Fortune 500 clients. link

Summer Software Intern **IoT Security**
Location: Icon Laboratory, Iowa *May 2017 – Jul 2017*

- Designed and implemented a security module for IoT devices, incorporating obfuscation and secure key-retrieval mechanisms to improve data privacy and system reliability.
- Integrated lightweight cryptographic techniques tailored for resource-constrained embedded IoT environments.

Leadership & Service

- **Ethics Reviewer** for the Dataset and Benchmarking Track, NeurIPS 2024 (USA).
- **Integration Testing Lead** across multiple Kingland projects, improving code coverage by **80%** and coordinating framework adoption across engineering teams in the US and China, 2020 (USA).
- **Poster Presenter and Participant** in the MoML Conference & Hackathon, Valence Labs & Mila, 2024 (Canada).
- **Poster Presenter** at the OSU Research Showcase, Oregon State University, 2024 (USA).
- **Founder & Organizer** for Explainable AI reading group at Oregon State University, 2024 (USA)
- **Member & Speaker** at the Pervasive Personalized Intelligence Center (PPIC), 2023 (USA).
- **Participant** in the AgAID Hackathon, 2022 (USA).
- **Mentor** to three junior engineers, providing structured guidance in UI development and software testing, (USA).
- **Honourable Mention** at the Kingland Poster Presentation, Iowa State University (USA).
- **Chairperson** for student chapter of Computer Society of India (India).

Skills

Programming: Python, SQL, Java, C/C++, C#
Machine Learning: Statistical Modeling, Experiment Design, Model Evaluation, Explainable AI
Frameworks & Tools: PyTorch, TensorFlow, Keras, Pandas, NumPy, Scikit-learn
Systems: AWS, Docker, Terraform, Jenkins, Git, CI/CD Pipelines
Data: Large-scale data processing, feature engineering, analytics pipelines